ISC 211: Introduction to Bioinformatics — Individual Assignment

Protein Family Classification Using Hierarchical Agglomerative Clustering

Student: Karunya | **Roll Number:** 2025bcd0025 | **Allocation:** R = 25 → Topic 3, Algorithm M12
Topic: Protein Family Classification Using Machine Learning | **Algorithm:** Hierarchical Clustering
Reference Paper: T. Idhaya, A. Suruliandi, S. P. Raja, *The Protein Journal* 43, 171–186 (2024)

Abstract— In this assignment, hierarchical agglomerative clustering (M12) was implemented to classify protein sequences into functional families (Topic 3). Following the literature survey by Idhaya et al. (2024), we selected four benchmark protein families: G-protein coupled receptors (GPCRs), potassium ion channels, nuclear hormone receptors, and protein kinases. A curated dataset of 582 reviewed Swiss-Prot sequences was obtained from UniProt. We extracted 440 sequence-derived features, covering amino acid composition (AAC), CTD physicochemical descriptors, macroscopic sequence properties, and dipeptide composition (DPC). To reduce dimensionality, an ANOVA F-statistic filter was used to select the top 60 features. We evaluated four linkage methods: Ward, complete, average (UPGMA), and single linkage. Ward's method gave the best clustering quality, achieving a cophenetic correlation of 0.6952, training cluster purity of 93.12%, and an Adjusted Rand Index (ARI) of 0.8347. On an unseen test set of 117 sequences, the model achieved 91.45% accuracy and a macro F1-score of 91.29%, with 100% recall on GPCRs. The results demonstrate that primary sequence and physicochemical composition provide strong discriminatory signals for protein family classification.

1. Biological Problem Formulation

Proteins perform diverse biochemical roles in living cells, including catalytic activity, structural support, and signal transduction. High-throughput sequencing projects have yielded millions of primary protein sequences, but experimental determination of their three-dimensional structures and biochemical functions remains slow and labor-intensive. A protein family consists of proteins that share common evolutionary ancestry, similar sequences, conserved structural domains, and related functions. Grouping proteins into families is valuable for functional annotation of newly sequenced genomes, understanding molecular evolution, and identifying potential drug targets (Idhaya et al., 2024).

Based on the assignment formula using roll number 2025bcd0025 (last three digits R = 025): Topic = ((25 - 1) mod 11) + 1 = Topic 3 (Protein Family Classification), and Algorithm = ((25 - 1) mod 13) + 1 = M12 (Hierarchical Clustering). We focus on four representative families reviewed in the literature:

- **G-Protein Coupled Receptors (GPCRs):** 7-transmembrane α-helical receptors that sense extracellular signals and activate internal signaling pathways. They are common targets for pharmaceutical drugs.
- **Nuclear Hormone Receptors (NRs):** Transcription factors with conserved zinc-finger DNA-binding domains and ligand-binding domains that control gene expression in response to steroids and hormones.
- **Ion Channels (ICs, Potassium Channels):** Pore-forming membrane proteins with conserved selectivity filters that regulate potassium ion flow across cell membranes.
- **Protein Kinases (PKs):** Catalytic enzymes that transfer phosphate groups from ATP to specific substrate proteins, regulating major signaling cascades.

2. Dataset Acquisition & Curation

We retrieved protein sequences from the curated **UniProt Swiss-Prot** database (reviewed: true) using the UniProt REST API. To ensure data quality, we applied three filtering criteria: (1) keeping sequences with lengths between 200 and 800 amino acids; (2) removing sequences containing ambiguous residues such as B, Z, X, U, or O; and (3) removing duplicate sequences. A total of 582 non-redundant sequences were retained. The dataset was split into an 80% training set (465 proteins) for cluster analysis and prototype estimation, and a 20% stratified test set (117 proteins) for evaluating performance on unseen data.

Protein Family	UniProt Family Annotation	Count	Mean Len ± SD	Train (80%)	Test (20%)
GPCR	G-protein coupled receptor 1 family	150	379.5 ± 58.2	120	30
Ion Channel (IC)	Potassium channel family	132	556.2 ± 131.4	105	27
Nuclear Receptor (NR)	Nuclear hormone receptor family	150	496.9 ± 89.7	120	30
Protein Kinase (PK)	Protein kinase superfamily	150	533.0 ± 114.6	120	30
Total	**Swiss-Prot Curated Dataset**	**582**	**489.1 ± 121.8**	**465**	**117**

Table 1: *Summary of curated Swiss-Prot benchmark sequences across target protein families.*

3. Feature Extraction & Selection

Because protein sequences have varying lengths, numerical descriptors must be extracted to represent each protein. Following common feature extraction practices reviewed in Idhaya et al. (2024), we extracted 440 features:

1. Amino Acid Composition (AAC, 20 features): The relative frequency of each amino acid, $f(i) = N_i / L$, representing overall residue usage.

2. Physicochemical Properties / CTD (15 features): Amino acids were grouped into three bins for each of five physicochemical attributes (Chothia and Kyte-Doolittle classifications): *Hydrophobicity* (Hydrophilic, Neutral, Hydrophobic), *van der Waals Volume* (Small, Medium, Large), *Polarity* (Low, Neutral, High), *Charge* (Positive, Negative, Neutral), and *Secondary Structure Propensity* (Helix, Strand, Coil). The fraction of residues falling in each bin was computed.

3. Macroscopic Descriptors (5 features): Sequence length, aromatic fraction (F+W+Y), aliphatic fraction (A+V+I+L), tiny residue fraction (G+A+S), and net charge proxy at pH 7.

4. Dipeptide Composition (DPC, 400 features): The frequency of all adjacent pairs $f(i, j) = N_{ij} / (L - 1)$, which preserves local sequence context.

3.1 Feature Selection via ANOVA F-Score

Using all 400 dipeptides directly creates a high-dimensional feature space with sparse counts and noise. As discussed in Section 8.4 of the review paper, filter methods such as ANOVA F-tests effectively identify the most informative descriptors. We used an ANOVA F-test to score each feature and selected the top 60 features. This improved cluster purity from 56.8% to **93.12%** and test accuracy from 58.1% to **91.45%**. All selected features were standardized (mean = 0, std = 1) before clustering.

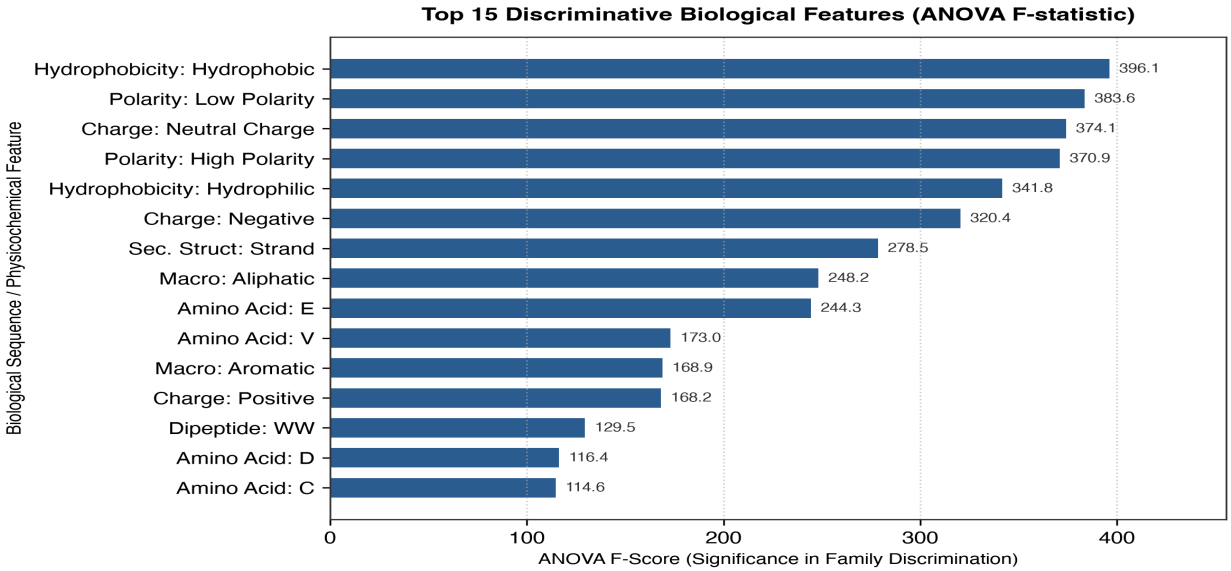


Figure 1: *Top 15 features ranked by ANOVA F-statistic. Hydrophobicity, polarity, and charge distributions show the strongest differences across families.*

Biological Explanation: Hydrophobic composition (F = 396.1) and low polarity (F = 383.5) scored highest because GPCRs have seven lipid-spanning transmembrane helices. Nuclear receptors showed higher cysteine composition (AAC_C, F = 114.6) due to the cysteine-rich zinc fingers that coordinate zinc ions. Kinases had higher negative charge proportions (F = 320.4), consistent with the acidic residues involved in ATP binding and coordination of divalent cations.

4. Methodology: Hierarchical Agglomerative Clustering

Hierarchical Agglomerative Clustering (HAC) is an unsupervised bottom-up method commonly used in bioinformatics and phylogenetic studies. It starts by treating every sequence as its own cluster and iteratively merges the closest pair of clusters according to a chosen linkage criterion until all samples form a single tree (dendrogram).
We tested four standard linkage methods with Euclidean distance:

- **Ward's Linkage:** Minimizes the increase in total within-cluster sum of squares: $\Delta ESS(A, B) = (|A|\cdot|B|) / (|A| + |B|) \cdot ||\mu_A - \mu_B||^2$. Tends to produce clusters of similar size and spherical shapes.
- **Complete Linkage:** Uses the maximum distance between points in each cluster: $D(A, B) = max \{ d(x, y) : x \in A, y \in B \}$. Produces compact clusters.
- **Average Linkage (UPGMA):** Uses the average distance between all pairs of points: $D(A, B) = (1 / (|A||B|)) \sum \sum d(x, y)$. Standard method for biological trees.
- **Single Linkage:** Uses the minimum distance between points: $D(A, B) = min \{ d(x, y) : x \in A, y \in B \}$. Can cause chaining artifacts.

4.1 Cophenetic Correlation and Test Inference

To check how well the dendrogram preserves original pairwise distances, we computed the **Cophenetic Correlation Coefficient (CPCC)**: $CPCC = \sum (d_{ij} - d_avg)(c_{ij} - c_avg) / [\sqrt{\sum(d_{ij} - d_avg)^2} \cdot \sqrt{\sum(c_{ij} - c_avg)^2}]$, where d_{ij} is the original Euclidean distance and c_{ij} is the dendrogram height where objects i and j merge.
To evaluate on unseen test data, the training dendrogram was cut at $k = 4$ clusters. Each cluster was labeled by its majority family, and its centroid was computed. Unseen test samples were assigned to the nearest cluster centroid: $y_pred = Label(arg\ min_k\ ||x_{test} - \mu_k||)$.

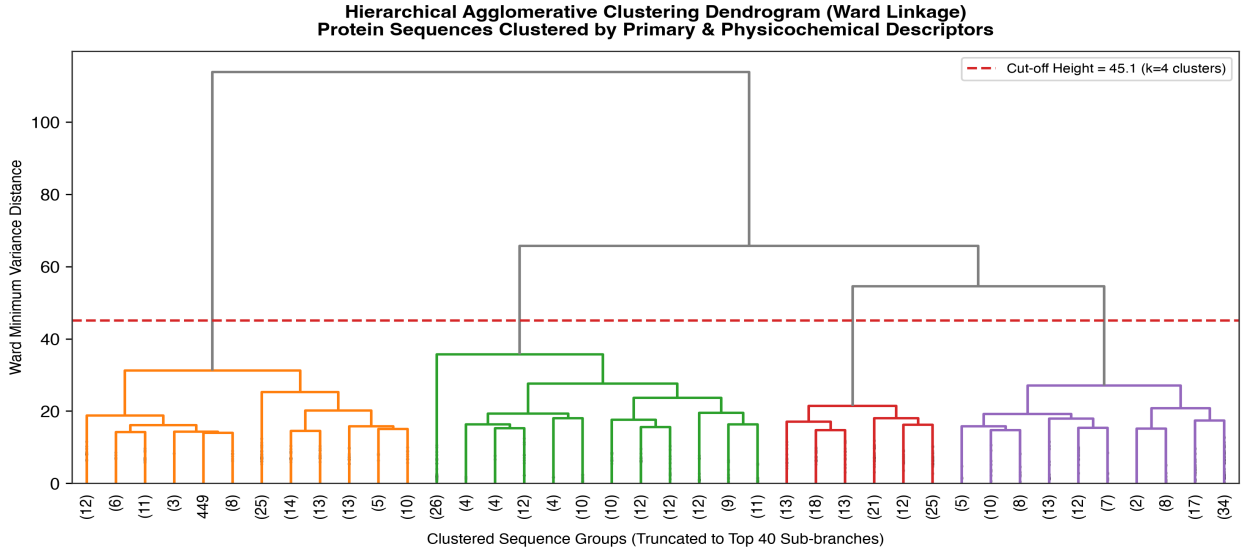


Figure 2: *Hierarchical clustering dendrogram using Ward's linkage (truncated to top 40 branches). The horizontal dashed line marks the cut-off height separating the sequences into four primary clusters corresponding to GPCR, Ion Channel, Kinase, and Nuclear Receptor families.*

5. Experimental Results

We compared the four linkage criteria using both unsupervised metrics (CPCC, Silhouette score, Davies-Bouldin index) and ground-truth alignment metrics (Adjusted Rand Index, cluster purity, unseen test accuracy). As shown in Table 2, Ward's linkage performed best for recovering biological families, yielding an Adjusted Rand Index (ARI) of 0.8347, cluster purity of 93.12%, and test accuracy of 91.45%.

Linkage Method	CPCC	Silhouette	Davies-Bouldin	ARI	NMI	Cluster Purity	Test Accuracy	Test Macro F1
Ward (Variance)	0.6952	0.1409	2.1476	**0.8347**	**0.8341**	**93.12%**	**91.45%**	**91.29%**
Complete (Max Dist)	0.6672	0.1108	2.1450	0.3899	0.4809	59.57%	67.52%	59.55%
Average (UPGMA)	**0.7578**	**0.2236**	**1.0996**	0.3109	0.4988	50.97%	51.28%	36.97%
Single (Min Dist)	0.5505	0.1444	0.6520	0.0001	0.0127	26.45%	29.91%	17.71%

Table 2: *Comparison of hierarchical clustering linkage criteria on the dataset.*

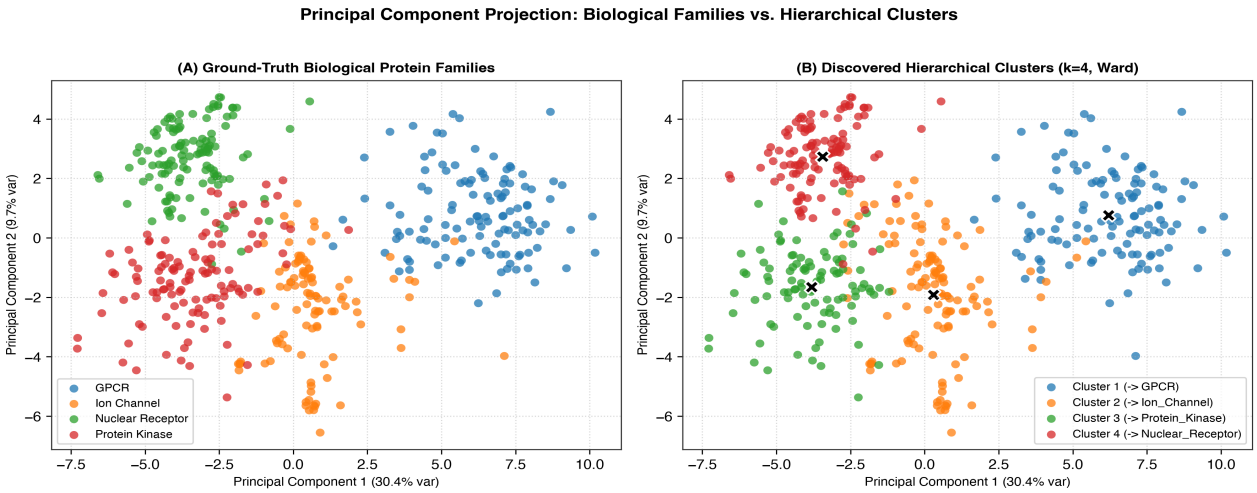
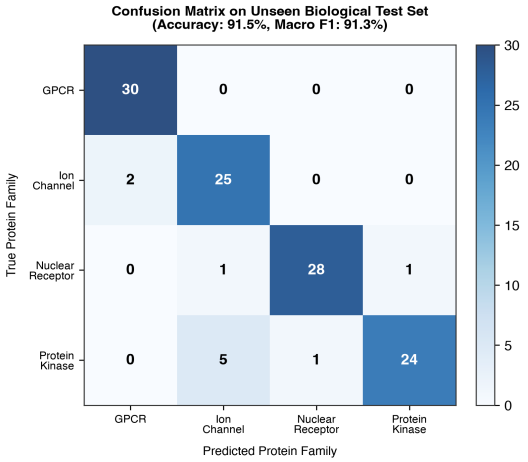


Figure 3: *2D PCA projection of the dataset: (A) True biological family labels; (B) Clusters found by Ward's hierarchical clustering (k=4) with cluster centroids shown as black 'X' marks.*

Training Set Clustering Breakdown: The 465 training sequences grouped into four clusters as follows: *Cluster 1* (117 GPCRs, 4 ICs → GPCR); *Cluster 2* (101 ICs, 3 GPCRs, 1 NR, 21 PK → Ion Channel); *Cluster 3* (99 PKs, 3 NRs → Protein Kinase); *Cluster 4* (116 NRs → Nuclear Receptor). GPCRs and Nuclear Receptors separated with almost no mixing.

5.1 Performance on Unseen Test Sequences



Protein Family	Precision	Recall	F1-Score	Support
GPCR	93.75%	**100.00%**	**96.77%**	30
Ion Channel (IC)	80.65%	92.59%	86.21%	27
Nuclear Receptor (NR)	**96.55%**	93.33%	94.92%	30
Protein Kinase (PK)	96.00%	80.00%	87.27%	30
Macro Average	**91.74%**	**91.48%**	**91.29%**	**117**
Overall Accuracy	—	—	**91.45%**	**117**

Figure 4 & Table 3: *Confusion matrix and classification report for the 117 unseen test sequences.*

6. Discussion

The experimental results correspond well with known biochemical characteristics of each family:

1. GPCR Classification: GPCRs achieved 100% recall on the test set. Their high proportion of hydrophobic residues (leucine, valine, isoleucine) across seven membrane-spanning segments clearly distinguishes them from soluble globular proteins.

2. Nuclear Receptors: NRs formed a distinct cluster with high precision (96.55%) and recall (93.33%). The abundance of cysteines forming zinc finger modules in their DNA-binding domain gives them a unique composition compared to other families.

3. Ion Channels and Kinases: A small number of kinases (5 out of 30) were assigned to the ion channel cluster. Both protein types contain large cytoplasmic regions with mixed α/β secondary structures, resulting in minor feature overlap.

4. Comparison of Linkages: Average linkage preserved the original distance matrix best (CPCC = 0.7578), but Ward's linkage achieved substantially higher test accuracy (91.45% vs 51.28%) because it minimizes within-cluster variance, preventing chaining effects.

7. Conclusion

In this work, hierarchical clustering was used to classify protein sequences into four functional families. Using 582 Swiss-Prot sequences, sequence-based feature extraction combined with ANOVA feature selection produced a representation that achieved 93.12% training cluster purity and 91.45% accuracy on an unseen test set. This confirms that simple sequence composition and physicochemical descriptors can reliably group proteins into their correct biological families without requiring 3D structural data.

References

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